

PROGRAMMABLE ASSETS: THE FUTURE OF FINANCE



Post internet, the emergence of blockchain has been the most remarkable accomplishment.

During the last decade, it has found widespread applications in the global economy across several segments of services and manufacturing. It led to the birth of virtual digital assets and opened a new methodology of transactions, like cryptocurrency, stable coins, central-bank digital currency, non-fungible tokens, decentralized financing, gaming tokens – all now entering a mature phrase.

Parallel developments in AI, particularly after the success of Alpha-Zero in 2017, increased the pace of machine-assisted, intelligent decision making. All these new formats of transactions had a unique feature – they were self-regulating. With the openness of public ledger, these provided trustworthiness, and with encryption and blockchain, the transactions also provided anonymity. Leveraging such obvious advantages, the concept of smart contracts was devised.

“One of the reasons I’m excited about blockchains is because I see them as a way to return us to predictable platforms”, remarked Chris Dixon, Venture Capitalist at Andreessen Horowitz, “where entrepreneurs and creators can build direct relationships with their audiences. This is what the internet was supposed to be.” Smart contracts based on blockchain soon started changing the rules of the game. Based on the concept of self-operating agreements, they have now become a potent tool to perform the basic purpose of any contract: verify the conditions that fulfil the laid down terms, facilitate the processing of the agreements, and finally enforce the terms of the contract between the interested parties. All this happens without any manual assistance, substantially reducing the procedural bottlenecks and the time taken to make a considered decision.

The limitation of smart contracts is their rigidity. Once deployed, they become fixed and if any amendments need to be made, these need fresh deployment. But the new generation of instrument – that of programmable assets, while addressing these limitations, have added many more advantages. Utilizing the same blockchain technology, programmable assets have greater flexibility, wider scope and much fluid adaptability

due to its structure.

Programmable assets can primarily be seen as currency or tokens of future finance. These are digital objects, with in-built rules of transaction. They have enormous flexibility to interact with other assets individually.

Unlike smart contracts, these assets are modular and can be reused. This flexibility imparts to them the character of a digital currency. It also, by implication, allows seamless inheritance. Another advantage is the security. The data of the user is encapsulated in each of the token, and with appropriate encryption, limits the possibilities of unauthorized access.

The most advantageous element of the programmable assets is their meta-class architecture. The transactions offer a higher level of security across the nodes. While reducing vulnerability, programmable assets also allow higher flexibility to update and make changes without hampering the flow across the chain.

The three models of programmable digital currencies identified by Monetary Authority of Singapore are programmable money, that can be used with pre-defined protocols; programmable payments, like dividends based on multitude of parameters; and purpose-bound money to facilitate P2P transfers. One of the early successful programmable assets is JP Morgan Coin, which has been in vogue since 2020, with systematic adaptations. Several initiatives applying the programmable asset class are emerging in the global financial horizon. BUIDL, the Ethereum based fund of BlackRock, launched in March last year is backed by short-term U.S. Treasuries and has \$520 million of assets.

The applications of programmable assets are not limited to financial transactions alone. It has expanded to envelope several other segments successfully including supply-chain management, healthcare, and research and development, to name a few. Its trackability, traceability, scalability and speed has also made it a very adaptive tool for vehicle leasing, vehicle management and performance management in the automotive sector.

Such a centralized and dynamically programmable transaction platform offers enormous opportunities.

“As the value goes up, heads start to swivel and skeptics begin to soften” remarks Adam B. Levine, CEO of Tokenly. The regulatory framework for these new technologies is gradually evolving. There are some inherent challenges though. The foremost is accountability. As in most emerging technologies, it becomes difficult to determine the liability in the event of an unforeseen or unintended malfunction. Internet has created a borderless society. While it has facilitated phenomenal growth of economies, it also calls for the harmonization of regulatory standards. Realizing the potential threats, recommendations 15 and 16 of the global body, the Financial Action Task Force (FATF) prescribe for close monitoring of virtual assets. The Targeted Update on Implementation of the FATF Standards on Virtual Assets, 2024 lays down elaborate guidelines and plan of action.

To achieve the full advantage of programmable assets, and to take the global economy to more inclusive heights, a much greater cooperation among we, the dwellers of the global village, created by internet, is becoming the precondition. Ultimately, regulation is not just a safeguard but an enabler, creating the foundation for programmable assets to scale responsibly, integrate seamlessly with traditional systems, and unlock their full potential in transforming industries from finance to healthcare and governance.



Jayant Misra

Ex- IRS, Financial Regulatory Expert
Ex-Chief Commissioner CBIC